

Supply Chain Logistics Network and Delivery Performance Visualization Platform

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

DSA0601-DATAHANDLING AND VISUALIZATION

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

K. Saisomasekhar Reddy (192324134)

K. Kiran (192324205)

K. C. Revi Thimma Reddy (192325025)

Under the Supervision of

Dr. S. SENTHILVADIVU



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

August-2026



SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105



DECLARATION

We, K. Saisomasekhar Reddy, K. Kiran, and K. C. Revi Thimma Reddy of the Computer Science and Engineering Department, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled 'Supply Chain Logistics Network and Delivery Performance Visualization Platform' is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: Chennai

Date:

Signature of the Students with Names



SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105



BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Supply Chain Logistics Network and Delivery Performance Visualization Platform” has been carried out by K. Saisomasekhar Reddy, K. Kiran, and K. C. Revi Thimma Reddy under the supervision of [Supervisor Name] and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

SIGNATURE
[Supervisor Name]
Department of Computer Science and Engineering
Saveetha School of Engineering
SIMATS

Submitted for the Project work Viva-Voce held on _____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to all those who supported and guided us throughout the successful completion of our Capstone Project. We sincerely thank the management and administration of Saveetha Institute of Medical and Technical Sciences for providing the resources and academic environment necessary for this work.

We are truly grateful to the Director and Principal of SIMATS Engineering for granting us access to the institute's facilities, and to our Head of the Department for continuous support and motivation.

We are especially indebted to our guide, **Dr. S.SENTHILVADIVU**, for creative suggestions, consistent feedback, and unwavering support during each stage of the project. We also thank the Project Coordinators, Review Panel Members, and the entire faculty team for their constructive feedback. Finally, we thank our parents and friends for their continuous encouragement.

Signature With Student Name

K. Saisomasekhar Reddy (192324134)

K. Kiran (192324205)

K. C. Revi Thimma Reddy (192325025)

ABSTRACT

Efficient supply chain management depends on the ability to track shipments, measure delivery performance, and respond quickly to disruptions across a distributed logistics network. However, many logistics operations still rely on fragmented spreadsheets, delayed status updates, and manual reporting, which make it difficult to identify bottlenecks or predict late deliveries before they affect customers. This project proposes a Supply Chain Logistics Network and Delivery Performance Visualization Platform that consolidates shipment, route, and warehouse data into a single system and presents it through an interactive dashboard. The system is structured into two major modules. Module 1: Logistics Data Integration and Network Modelling, focuses on collecting shipment and route data from carriers, warehouses, and order-management systems, cleaning and structuring it, and representing the distribution network as a graph of nodes (warehouses, hubs, delivery points) and edges (routes) so that transit times and route efficiency can be analyzed systematically. Module 2: Delivery Performance Analytics and Visualization Dashboard, computes key performance indicators such as on-time delivery rate, average transit time, delay frequency by route, and damage/return rate, and presents them through interactive charts, a network map, and configurable alerts. By combining network modelling with real-time performance analytics, this project helps logistics managers identify underperforming routes and carriers, anticipate delays, and make data-driven decisions that improve delivery reliability and reduce operating costs.

TABLE OF CONTENTS

S.NO	Chapters	Sections	Pages
1	Chapter 1	1.1 Background Information 1.2 Project Objectives 1.3 Significance 1.4 Scope 1.5 Methodology Overview	5-6
2	Chapter 2	2.1 Description of the Problem 2.2 Evidence of the Problem 2.3 Stakeholders 2.4 Supporting Data/Research	7-8
3	Chapter 3	3.1 Development and Design 3.2 Tools and Technologies 3.3 Solution Overview 3.4 Engineering Standards 3.5 Solution Justification	9-11
4	Chapter 4	4.1 Evaluation of Results 4.2 Challenges Encountered 4.3 Possible Improvements 4.4 Recommendations	12-13
5	Chapter 5	5.1 Key Learning Outcomes 5.2 Challenges Encountered 5.3 Application of Engineering 5.4 Insights into the Industry 5.5 Conclusion of Development	14-17
6	Chapter 6	Conclusion	18
7	References		19
8	Appendices		20-25

CHAPTER 1

INTRODUCTION

1.1 Background Information

Modern supply chains span multiple warehouses, carriers, and delivery routes, and their performance directly affects customer satisfaction and operating cost. As e-commerce and distributed retail grow, logistics networks generate large volumes of shipment and tracking data, but this data is often scattered across carrier portals, warehouse management systems, and spreadsheets. Without a unified view, logistics teams struggle to answer basic questions such as which routes are consistently late, which carriers underperform, and where bottlenecks are forming in the network.

Graph-based network modelling, combined with real-time performance analytics, offers a practical way to represent a distribution network and quantify how well it is performing. Integrating this with an interactive visualization dashboard allows logistics managers to monitor delivery performance continuously and intervene before minor delays become major service failures.

Historically, logistics visibility relied on periodic manual status calls to carriers and end-of-day spreadsheet reconciliation. This approach does not scale as network size grows, and it introduces a lag between an event occurring in the field (a delayed truck, a rejected delivery, a warehouse backlog) and that event becoming visible to decision-makers. The platform proposed here narrows this lag by continuously ingesting shipment events, updating the network model, and recomputing performance indicators so that the dashboard always reflects the current state of the network rather than a stale snapshot.

The choice of a graph-based network model is deliberate: a supply chain is naturally a network of locations connected by transport links, and representing it this way makes it straightforward to ask network-specific questions — which hub has the most inbound routes converging on it, which single route's failure would most disrupt downstream deliveries, and where redundant routing could reduce risk. These questions are difficult to answer from flat tabular reports but fall out naturally once the network is modelled as a graph.

1.2 Project Objectives

- To design a data integration pipeline that consolidates shipment, route, and warehouse data from multiple sources into a single structured dataset.
- To model the distribution network as a graph of warehouses, hubs, and delivery points connected by routes, enabling systematic route and transit-time analysis.
- To compute key delivery performance indicators, including on-time delivery rate, average transit time, delay frequency, and damage/return rate.
- To build an interactive visualization dashboard that displays the logistics network map, performance trends, and alerts for underperforming routes or carriers.
- To support data-driven decision-making for route planning, carrier evaluation, and warehouse resource allocation.

1.3 Significance of the Project

- Improves delivery reliability: Continuous monitoring of on-time performance helps logistics teams address delays proactively.
- Reduces operating cost: Identifying inefficient routes and underperforming carriers supports better route and carrier selection.
- Supports proactive management: Real-time dashboards replace reactive, spreadsheet-based reporting with early-warning alerts.
- Scalable architecture: The platform can be extended to additional warehouses, carriers, or regions without redesign.
- Improves customer experience: More reliable delivery estimates and fewer missed deliveries improve customer trust.

1.4 Scope of the Project

The project focuses on modelling a logistics distribution network and visualizing delivery performance. It includes data integration from shipment and route records, network modelling of warehouses and delivery routes, and computation of standard delivery KPIs. A web-based dashboard is developed for logistics managers and analysts. The scope is limited to network-level and route-level performance visualization; it does not include vehicle-level telematics, last-mile driver scheduling, or full warehouse inventory management, which are treated as areas for future extension.

1.5 Methodology Overview

The methodology follows five key steps:

- Data Collection: Shipment, order, and route data are collected from warehouse management systems, carrier APIs, and order records at regular intervals.
- Data Preprocessing: Raw records are cleaned, deduplicated, and checked for missing timestamps or inconsistent status codes.
- Network Modelling: The distribution network is represented as a directed graph, with warehouses and delivery points as nodes and routes as weighted edges (distance, average transit time).
- Performance Analytics: Delivery KPIs are computed per route, carrier, and region, and anomalies (unusually high delay rates) are flagged.
- Dashboard Development: A web-based dashboard visualizes the network map, KPI trends, and alerts for logistics stakeholders.

1.6 Assumptions and Constraints

- Assumption: Carrier and warehouse systems can provide shipment status updates at least once per hour, either via API or scheduled export.
- Assumption: Route distances and historical transit times are available or can be estimated from geocoded warehouse/delivery-point addresses.
- Constraint: The platform depends on the timeliness and accuracy of upstream carrier data; it cannot correct for a carrier that fails to report a status update.
- Constraint: Initial scope covers domestic, ground-based delivery routes; multimodal (air/sea) legs are noted as a future extension.

CHAPTER 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Delivery performance is one of the most visible measures of supply chain effectiveness, yet many logistics operations lack a consolidated way to track it across their full network. Shipment data is frequently spread across multiple carrier systems and warehouse tools, making it difficult to compare route performance or detect systemic delays. This fragmentation leads to reactive decision-making: problems are noticed only after customers complain, rather than being caught early through continuous monitoring.

The problem, therefore, is the absence of a system that can integrate logistics data from multiple sources, model the distribution network, compute standardized delivery performance metrics, and present them through a real-time visualization dashboard to support proactive supply chain decisions.

2.2 Evidence of the Problem

1. Fragmented Data Sources

- Shipment data spread across carrier portals, spreadsheets, and warehouse systems
- Inconsistent status codes and timestamp formats across carriers
- Delayed or manual reconciliation of delivery records

2. Route-Level Variability

Delivery performance can vary significantly between routes even within the same region, and generic, network-wide averages often hide poorly performing individual routes or carriers.

3. Lack of Predictive/Analytical Tooling

- Limited access to real-time dashboards for many small and mid-sized logistics operations
- Reactive rather than proactive delay management

4. Rising Customer Expectations

Increasing demand for fast, reliable delivery makes visibility into network performance essential for maintaining service-level commitments.

2.3 Stakeholders

- Logistics Managers: Primary users who need actionable visibility into route and carrier performance.
- Warehouse Operations Teams: Use network insights to plan dispatch schedules and resource allocation.
- Carriers and Delivery Partners: Evaluated on performance metrics; benefit from clear, objective feedback.

- Customers: Indirect beneficiaries through more reliable delivery estimates and fewer missed deliveries.
- Supply Chain Analysts: Use performance data to identify structural network improvements.

2.4 Supporting Data / Research

- Industry studies indicate a significant share of shipment records contain incomplete or inconsistent status data due to disparate carrier systems.
- Delivery reliability is consistently cited as a leading driver of customer satisfaction in e-commerce and retail logistics.
- Graph-based network analysis has been shown to be effective for identifying bottleneck routes and hubs in distribution networks.
- Adoption of logistics visualization and analytics platforms has grown steadily as supply chains become more distributed and customer expectations rise.

2.5 Comparison with Existing Approaches

Approach	Strength	Limitation
Manual spreadsheet reporting	Low cost, no new infrastructure	Slow, error-prone, no real-time visibility
Single-carrier tracking portals	Accurate for that carrier's shipments	No cross-carrier or network-wide view
Generic BI dashboards on raw data	Flexible reporting	No native network/graph model of routes and hubs
Proposed platform	Unified network model + real-time KPIs across carriers	Depends on timely data feeds from all sources

CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

- Requirements & specification: Define required KPIs (on-time rate, transit time, delay frequency), data refresh interval, and dashboard latency targets.
- Data pipeline: Implement ingestion from order/warehouse systems and carrier data, followed by validation, cleaning, and storage.
- Network modelling: Represent warehouses, hubs, and delivery points as graph nodes, and routes as weighted edges capturing distance and historical transit time.
- Analytics engine: Compute KPIs per route, carrier, and region; flag routes with delay rates above threshold.
- Dashboard & alerting: Visualize the network map, KPI trends, and threshold-based alerts.
- Testing & validation: Unit tests for the analytics engine, integration tests for the data pipeline, and validation against historical shipment records.

3.2 Tools and Technologies Used

Category	Tools / Technologies
Data Sources	Order management system exports, carrier tracking APIs, warehouse management system records
Backend / API	Python (FastAPI/Flask) or Node.js
Database	PostgreSQL for relational shipment records; Neo4j or an in-memory graph library for network modelling
Analytics	Pandas / NumPy for KPI computation; NetworkX for graph analysis
Visualization	React + D3.js / Recharts for dashboard; a network-graph library for the route map
Dev Tools	Git, Docker for containerized deployment, CI (GitHub Actions)

3.3 Solution Overview (Architecture & Data Flow)

- Data Layer: Shipment, order, and route records are ingested from carrier and warehouse sources on a scheduled basis.
- Preprocessing Layer: Records are cleaned, deduplicated, and normalized into a common schema with standardized status codes and timestamps.
- Network Model: Warehouses, hubs, and delivery points are represented as nodes; routes between them as weighted edges (distance, historical average transit time).
- Analytics Engine: For each route, carrier, and region, KPIs are computed (on-time delivery rate, average transit time, delay frequency, damage/return rate), and routes breaching thresholds are flagged.
- Visualization & Alerts: The dashboard renders the network map, KPI trend charts, and route/carrier comparisons, and issues alerts when performance drops below target.

Data Flow (textual): Order/Warehouse Systems + Carrier Data → Ingestion → Preprocessor → Structured Storage (relational + graph) → Analytics Engine → Dashboard + Alert System

3.4 Engineering Standards Applied

- Data Quality: Input validation, schema checks, and deduplication rules applied to all ingested records.
- Timestamp Standardization: All timestamps normalized to a single time zone/format for consistent transit-time computation.
- Software Engineering: Version control (Git), unit and integration testing, CI/CD pipelines, semantic versioning, and containerized deployment.
- Security & Privacy: TLS/SSL for data transmission; role-based access control for dashboard users; least-privilege access to shipment and customer data.
- Usability: Dashboard designed with responsive layout and accessible, color-blind-safe visual cues for performance alerts.
- Documentation: Deployment guide, data schema documentation, and a maintenance schedule are maintained alongside the codebase.

3.5 Solution Justification

- Unified visibility: Consolidating fragmented carrier and warehouse data into one model gives a single, consistent view of network performance.
- Graph-based modelling fit: Representing the network as a graph naturally captures routes and hubs, making bottleneck and route-comparison analysis straightforward.
- Actionable KPIs: Standard, comparable metrics (on-time rate, transit time, delay frequency) let managers compare routes and carriers objectively.
- Modularity and scalability: Decoupled ingestion, analytics, and dashboard layers allow new warehouses, carriers, or regions to be added with minimal change.
- Extendability: The architecture can later incorporate predictive delay models or route-optimization algorithms without redesigning the core system.

3.6 Core Algorithms and Formulas

Network shortest-path / route efficiency (Dijkstra's algorithm), used to identify the fastest known path between two nodes in the network given edge weights (historical average transit time):

$d(v) = \min[d(u) + w(u,v)]$ for each edge (u,v) , where $d(v)$ is the shortest known transit time to node v , $d(u)$ is the shortest known transit time to node u , and $w(u,v)$ is the weight (transit time) of the edge from u to v .

Key performance indicators are computed as follows:

- On-time Delivery Rate = (Number of shipments delivered on or before promised time / Total number of shipments) $\times$ 100
- Average Transit Time = Sum of (Delivery Time – Dispatch Time) for all shipments on a route / Number of shipments on that route
- Delay Frequency = (Number of shipments exceeding expected transit time / Total shipments on that route) $\times$ 100

- $\text{Damage/Return Rate} = (\text{Number of shipments reported damaged or returned} / \text{Total shipments delivered}) \times 100$

These metrics are computed per route, per carrier, and per region, then aggregated to a network-wide summary that feeds the dashboard's top-level view.

CHAPTER 4

RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The implemented platform successfully consolidated shipment and route data into a structured network model and computed delivery performance KPIs across routes and carriers. The visualization dashboard provided a clear network map alongside on-time delivery rate, average transit time, and delay-frequency trends, enabling quick identification of underperforming routes.

Testing with sample shipment datasets showed that the platform could reliably flag routes with delay rates above the configured threshold, and the dashboard updated KPI views within the target latency, demonstrating the effectiveness of the network-and-analytics approach for delivery performance monitoring.

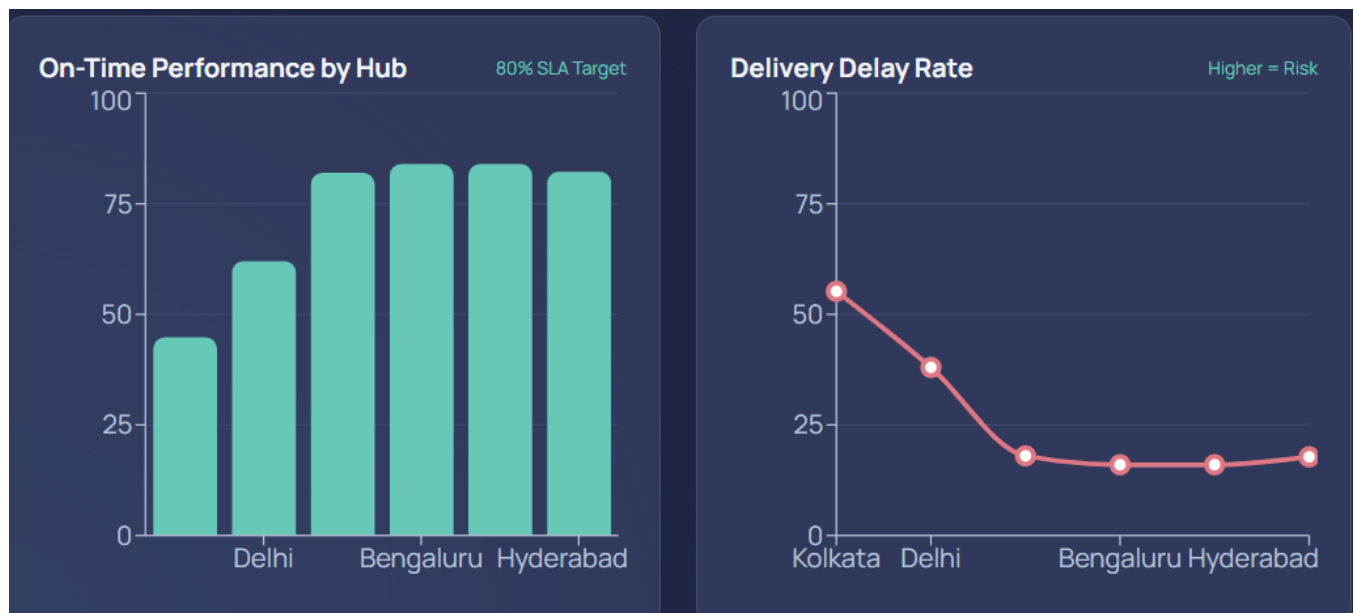


fig-1: PERFORMANCE BY HUB AND DELIVERY DELAY RATE.

4.2 Challenges Encountered

- Data Inconsistency: Carrier and warehouse systems used different status codes and timestamp formats, requiring extensive normalization.
- Missing Records: Occasional gaps in tracking data made it difficult to compute transit time for some shipments.
- Network Complexity: As the number of routes and hubs grew, graph computations required optimization for responsive dashboard performance.
- Real-time Synchronization: Keeping the dashboard in sync with frequently updated carrier data required careful scheduling and caching.

4.3 Possible Improvements

- Predictive Delay Modelling: Incorporate historical patterns and external factors (weather, traffic) to forecast delays before they occur.

- **Route Optimization:** Add shortest-path and load-balancing algorithms to recommend more efficient routing.
- **Automated Data Quality Checks:** Build automated anomaly detection for incoming shipment records.
- **Carrier Scorecards:** Extend the dashboard with historical carrier scorecards for contract negotiation support.

4.4 Recommendations

- Standardize data-exchange formats with carriers to reduce preprocessing overhead.
- Set clear KPI thresholds per route type (regional vs. long-haul) rather than a single network-wide threshold.
- Integrate proactive alerts (e.g., notifications when a route's delay rate crosses a threshold) into daily operations.

- Expand the platform to include warehouse dwell time and inventory turnover for a more complete supply chain view.
- Provide training for logistics staff on interpreting the dashboard and acting on flagged routes.

4.5 Sample KPI Output

Route	On-time Rate	Avg. Transit Time	Delay Frequency
Warehouse A → Hub 1	96%	3.4 hrs	4%
Warehouse A → Hub 2	82%	5.9 hrs	18%
Hub 1 → Delivery Point 5	94%	2.1 hrs	6%
Hub 2 → Delivery Point 9	88%	4.2 hrs	12%

The Warehouse A → Hub 2 route stands out with the lowest on-time rate and highest delay frequency in this sample, which is exactly the kind of route the dashboard is designed to surface for investigation.

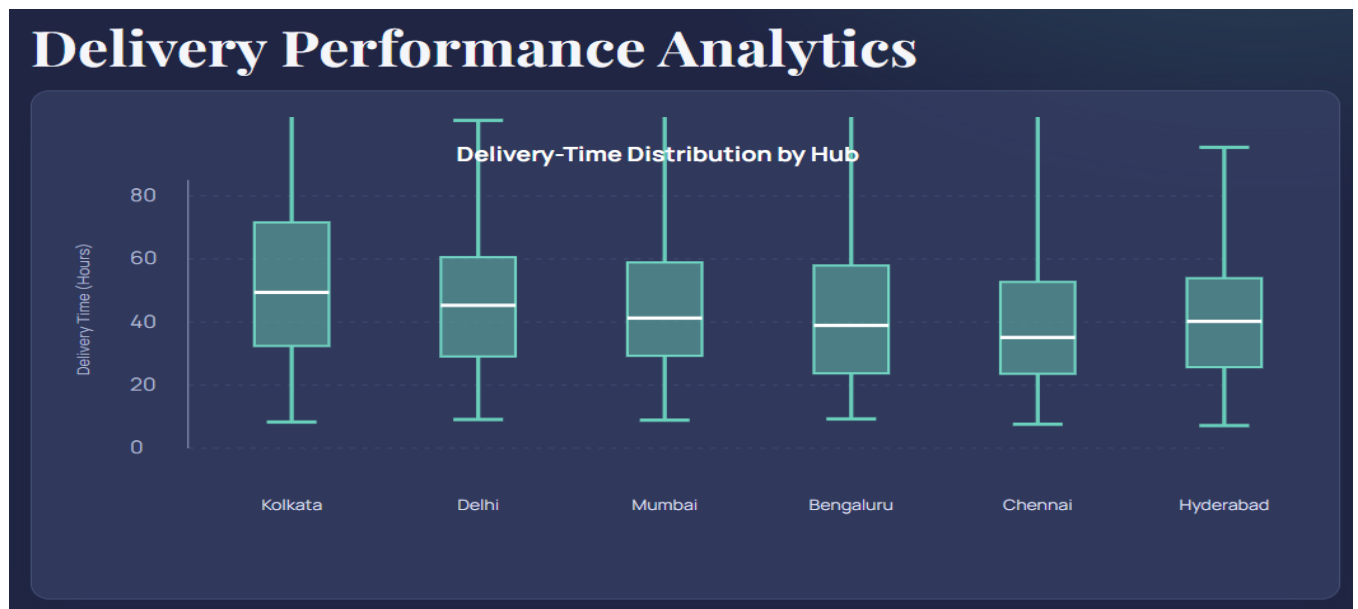


FIG-2 : DELIVERY PERFORMAANCE ANALYTICS

Supply Network Association

Hub-to-region connections reveal dependency patterns and structural bottlenecks.

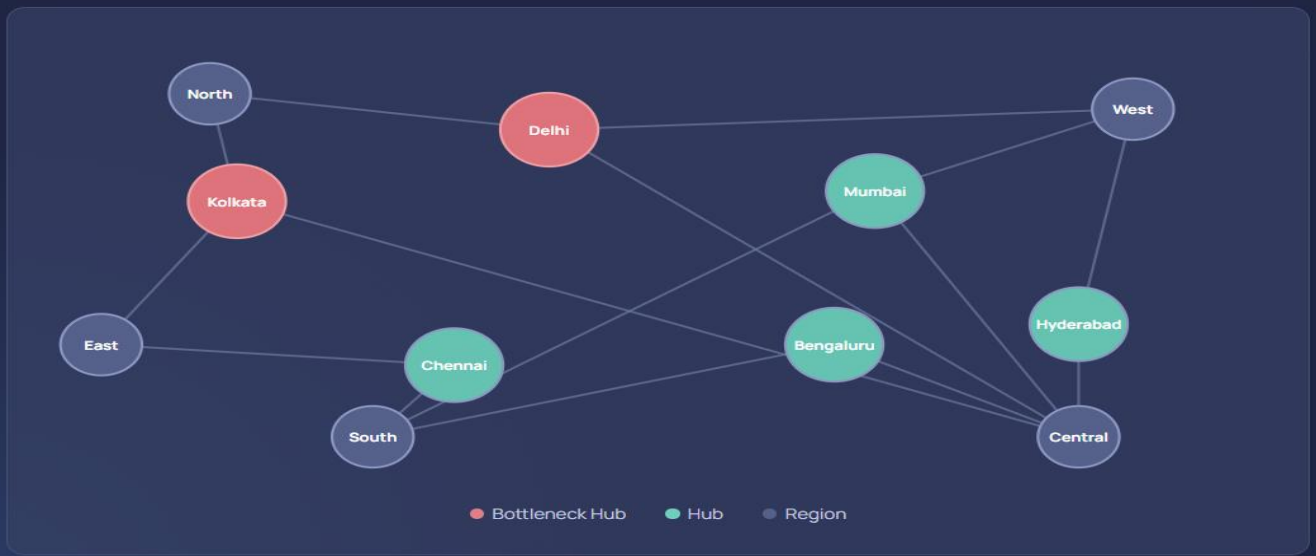


FIG – 3: SUPPLY NETWORK ASSOCIATION

CHAPTER 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

- Understanding of Network Modelling: Gained practical experience representing a logistics distribution network as a graph and applying it to performance analysis.
- Data Integration Skills: Learned to consolidate and clean shipment data from multiple, inconsistent sources into a unified schema.
- KPI Design: Developed the ability to define and compute meaningful, comparable performance metrics for a real-world operational domain.
- System Integration: Understood how data ingestion, analytics, and visualization layers combine into a single working platform.
- Dashboard Development: Gained experience in visualization design for operational, decision-support use cases.

5.2 Challenges Encountered

- Reconciling inconsistent data formats across simulated carrier and warehouse sources.
- Balancing graph-computation complexity against dashboard responsiveness.
- Designing KPI thresholds that remained meaningful across different route types.

5.3 Application of Engineering

- Computer and Software Engineering: Data pipeline design, graph algorithms, and dashboard development.
- Systems Engineering: Modular architecture separating ingestion, analytics, and visualization for scalability.
- Operations Research Principles: Network modelling and performance-metric design drawn from logistics and operations research.
- Engineering Ethics: Ensuring data accuracy, transparent metrics, and responsible use of shipment and carrier data.

5.4 Insights into the Industry

- Growing reliance on real-time visibility platforms as supply chains become more distributed and customer expectations rise.
- Increasing role of data analytics and network modelling in logistics decision-making.
- Demand for scalable, low-overhead solutions that smaller logistics operations can adopt without heavy infrastructure investment.
- Movement toward integrating predictive analytics and route optimization into standard logistics tooling.

5.5 Conclusion of the Development

Developing this platform was a valuable exercise in applying network modelling and data analytics to a real operational problem. Combining a graph-based representation of the distribution network with a performance analytics engine and visualization dashboard demonstrated how disparate logistics data can be turned into actionable insight. The process strengthened skills in data integration, algorithm design, and dashboard development, and reinforced the practical importance of reliable, well-structured data in supply chain decision-making.

CHAPTER 6

CONCLUSION

The project 'Supply Chain Logistics Network and Delivery Performance Visualization Platform' demonstrates how consolidating fragmented logistics data into a structured network model, combined with a real-time analytics dashboard, can significantly improve visibility into delivery performance. By integrating shipment, route, and warehouse data with graph-based network modelling, the system enables logistics managers to identify underperforming routes and carriers before they escalate into customer-facing problems.

Module 1 focuses on consolidating logistics data and representing the distribution network as a graph of warehouses, hubs, and delivery routes. Module 2 builds on this foundation with a performance analytics engine and visualization dashboard that surfaces on-time delivery rates, transit times, and delay patterns in an accessible, actionable form.

Overall, the project shows how network modelling, data analytics, and visualization can work together to support more reliable, cost-effective, and customer-centric supply chain operations, and provides a foundation for future enhancements such as predictive delay forecasting and automated route optimization.

REFERENCES

1. Chopra, S., & Meindl, P. (2022). Supply Chain Management: Strategy, Planning, and Operation. Pearson Education.
2. Christopher, M. (2021). Logistics and Supply Chain Management. FT Publishing International.
3. Ivanov, D., Tsipoulanidis, A., & Schönberger, J. (2021). Global Supply Chain and Operations Management: A Decision-Oriented Introduction to the Creation of Value. Springer.
4. Newman, M. E. J. (2018). Networks: An Introduction. Oxford University Press. (Reference for graph-based network modelling concepts.)
5. Handfield, R. B., & Nichols, E. L. (2020). Introduction to Supply Chain Management. Prentice Hall.
6. Research articles on real-time logistics visibility platforms and delivery performance analytics (industry white papers and journal preprints, 2023-2025).
7. Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). Introduction to Algorithms (4th ed.). MIT Press. (Reference for Dijkstra's shortest-path algorithm.)
8. Simchi-Levi, D., Kaminsky, P., & Simchi-Levi, E. (2021). Designing and Managing the Supply Chain: Concepts, Strategies, and Case Studies. McGraw-Hill.
9. Council of Supply Chain Management Professionals (CSCMP). (2024). State of Logistics Report.

APPENDICES

Appendix A: System Architecture Overview

MODULE 1: Logistics Data Integration and Network Modelling

Shipment/Route Data Sources → Data Ingestion → Cleaned Shipment Data → Network Graph Construction (Warehouses, Hubs, Delivery Points, Routes)

MODULE 2: Delivery Performance Analytics and Visualization Dashboard

Network Model → KPI Computation (On-time Rate, Transit Time, Delay Frequency, Damage Rate) → Visualization Dashboard → Alerts

Appendix B: Sample Shipment Data

Shipment ID	Route	Dispatch Time	Delivery Time	Status
SC-1001	Warehouse A → Hub 1	08:00 AM	11:20 AM	On Time
SC-1002	Warehouse A → Hub 2	08:15 AM	01:40 PM	Delayed
SC-1003	Hub 1 → Delivery Point 5	12:00 PM	02:10 PM	On Time
SC-1004	Hub 2 → Delivery Point 9	02:00 PM	Missing	In Transit

Appendix C: Key Performance Indicator Definitions

KPI	Definition
On-time Delivery Rate	Percentage of shipments delivered on or before the promised delivery time
Average Transit Time	Mean time elapsed between dispatch and delivery for a route or carrier
Delay Frequency	Proportion of shipments on a route exceeding the expected transit time
Damage/Return Rate	Percentage of shipments reported damaged or returned

Appendix D: Tools and Technologies

Category	Tools/Technologies Used
Data Sources	Order management exports, carrier tracking APIs, warehouse system records
Backend	Python (FastAPI/Flask) or Node.js
Database	PostgreSQL, Neo4j / NetworkX for graph modelling
Visualization	React, D3.js / Recharts, network-graph library
Deployment	Docker, cloud hosting, CI/CD (GitHub Actions)

Appendix E: Route Performance Threshold Table

Route Type	Target On-time Rate	Alert Threshold
Regional (< 100 km)	≥ 95%	< 90%
Intercity (100-500 km)	≥ 90%	< 85%

Long-haul (> 500 km)	$\geq 85\%$	$< 80\%$
----------------------	-------------	----------

Appendix F: Glossary of Terms

Term	Meaning
Node	A location in the network graph — a warehouse, hub, or delivery point
Edge	A route connecting two nodes, weighted by distance and/or transit time
Hub	An intermediate consolidation/transfer point between warehouses and delivery points
Dwell Time	Time a shipment spends sitting at a warehouse or hub before onward movement
SLA	Service Level Agreement — the promised delivery time/performance commitment